

DUT System

Linux + Native Drivers

Logical PCIe Interface

ECAM, BAR, INTx

INTx

DUT Memory

queues, descriptors, payload

FPGA Front-End

ECAM Shadow
+ BAR Routes

ECAM / BAR

Transaction Channel
+ Mailbox

ordered events

Interrupt Injector

pending level

DMA Windows

translated addresses

Host Software Back-End

Topology Manager
configuration + routes

Event Dispatcher +
Device-Semantic Adapters

physical control

Endpoint Contexts
state, binding, address map

Interrupt Manager
pending-state aggregation

completion cause

Physical Devices

Assigned Physical Device
NVMe SSD or
82580 NIC

device DMA

control / metadata

DMA payload / writebacks